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A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science

SECURIO

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Project Guide
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SDG

SDG 9: Industry, Innovation & Infrastructure — The app digitizes and automates residential society management through a mobile-based Android platform, replacing manual processes with innovative technology infrastructure using Firebase and Cloudinary.

SDG 11: Sustainable Cities & Communities — The app directly supports the goal of making human settlements inclusive, safe, and well-managed by digitizing visitor security, streamlining resident services, and improving overall urban housing society management.

1. Introduction

Problem Identified:

Current apartment security systems are manual and inefficient, with no real-time communication or proper visitor verification. They lack features like complaint management and maintenance billing, leading to poor record-keeping, low security, and inconvenience for residents.

Solution Proposed:

- Develop SECURIO, a digital visitor management system to replace manual registers.
- Enable real-time resident approval and maintain secure visitor logs in a centralized database.

2. Objectives

1. To prevent unauthorized access in residential buildings.
2. To enable real-time resident approval for visitor entry.
3. To digitize and automate the visitor entry process.
4. To maintain accurate visitor entry logs for future reference.

3. Scope

1. Can be applied in residential societies, apartments, and gated communities for digital visitor management.
2. Can be useful to security staff for registering and tracking visitor entries efficiently.
3. Can assist residents in approving or rejecting visitors through a mobile application.
4. Can be extended in future to include features like emergency alerts and advanced security monitoring.

4. Feature /Functionality

Feature 1: User Login & Authentication — Secure role-based login for Admin, Residents, and Security Guards using Firebase Authentication to ensure controlled and authorized access.

Feature 2: Visitor Registration & Approval — Security Guard registers visitor details including name and purpose of visit. Residents receive real-time notifications and can approve or reject entry before gate access is granted.

Feature 3: Maintenance Bill Management — Admin generates and issues monthly maintenance bills to residents. Residents can view bill details and track payment status digitally, reducing manual errors and missed payments.

Feature 4: Complaint Registration & Tracking — Residents can raise complaints, monitor their status in real-time, and receive updates once the issue is resolved by the admin.

Feature 5: Role-Based Access Control — Admin, Residents, and Security Guards are assigned separate permissions and dedicated dashboards, ensuring data privacy and controlled system access.

Feature 6: Cloud Data Storage — All data including visitor logs, maintenance bills, and complaints are securely stored using Firebase Firestore for reliability and scalability.

5. Outcome of Project

- **Digitized Visitor Management** — Successfully replaced manual register-based entry with a digital system where security guards log visitors and flat owners approve or reject entry in real-time.
- **Automated Maintenance Billing** — Maintenance bills are generated and tracked digitally, eliminating manual errors and ensuring timely payment communication between admin and residents.
- **Efficient Complaint Resolution** — Residents can raise and track complaints with live status updates, resulting in faster and more transparent resolution by admin.
- **Improved Society Security** — Role-based access and structured entry approval workflow significantly reduced unauthorized visitor entry and improved overall gate security.
- **Reduced Administrative Workload** — Automation of billing, complaints, and visitor management reduced manual effort and improved overall society management efficiency.

6. Technology Stack

Frontend

- **Platform** — Android (Mobile Application)
- **IDE** — Android Studio
- **Language** — Java / Kotlin
- **UI Design** — XML Layouts (Android Views)

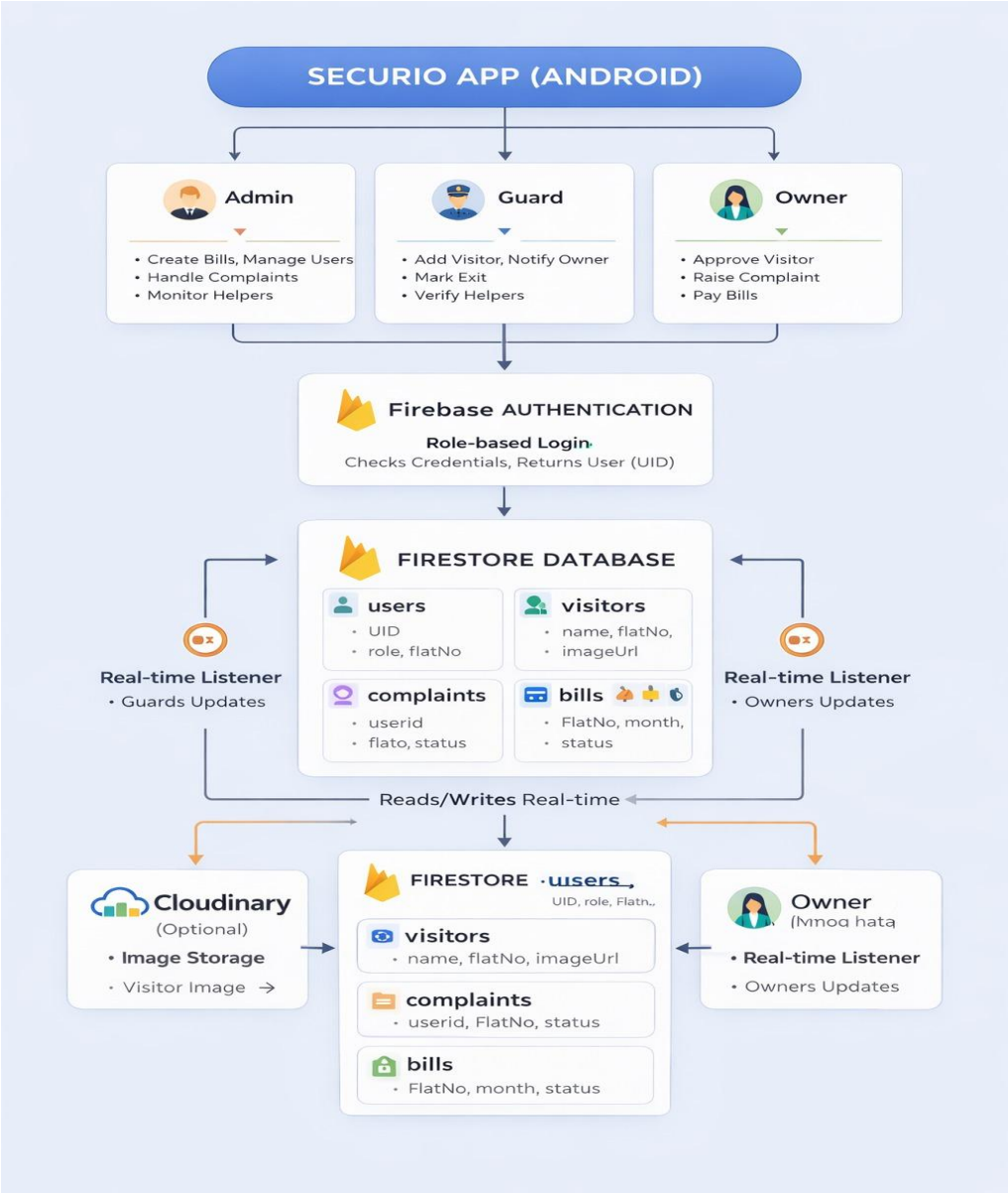
Backend

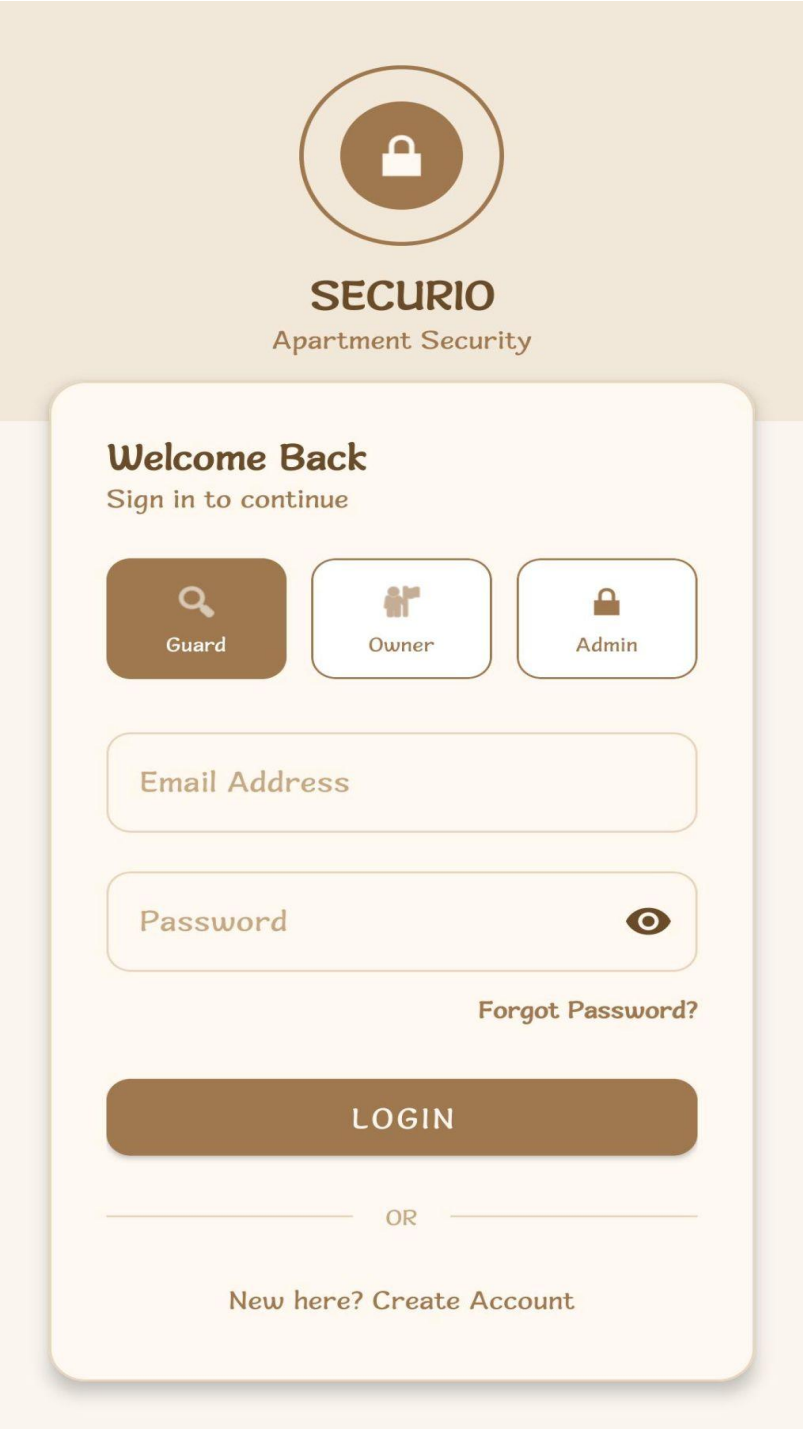
- **Database** — Firebase Firestore (Cloud-based data storage)
- **Authentication** — Firebase Authentication (Secure role-based login)
- **Real-Time Updates** — Firebase Realtime Database
- **Push Notifications** — Firebase Cloud Messaging (FCM)

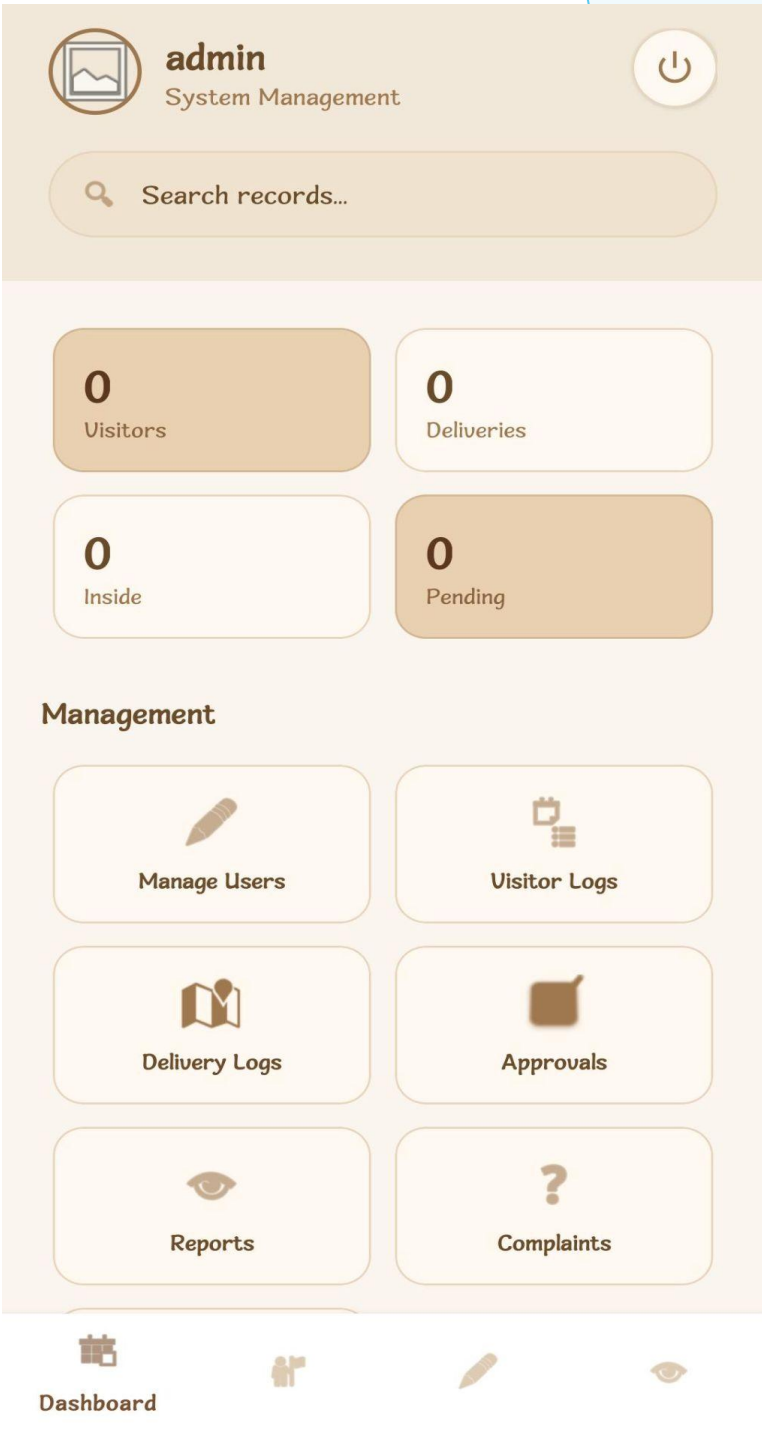
Storage

- **Image Storage** — Cloudinary (Store and manage profile and visitor images)

7. Block Diagram (if applicable)




SECURIO
Apartment Security
Welcome Back
Sign in to continue
Guard
Owner
Admin
Email Address
Password
Forgot Password?
LOGIN
OR
New here? Create Account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Visitors
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Deliveries
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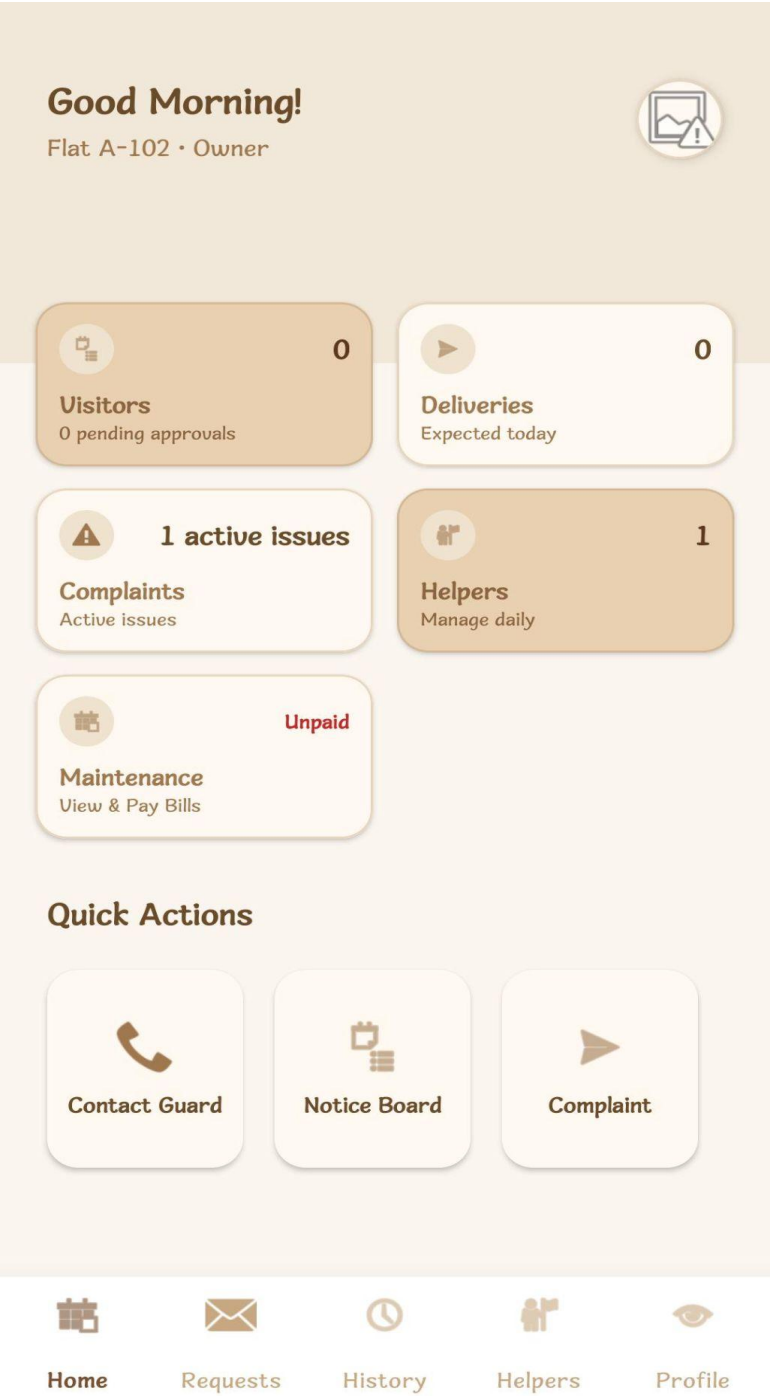
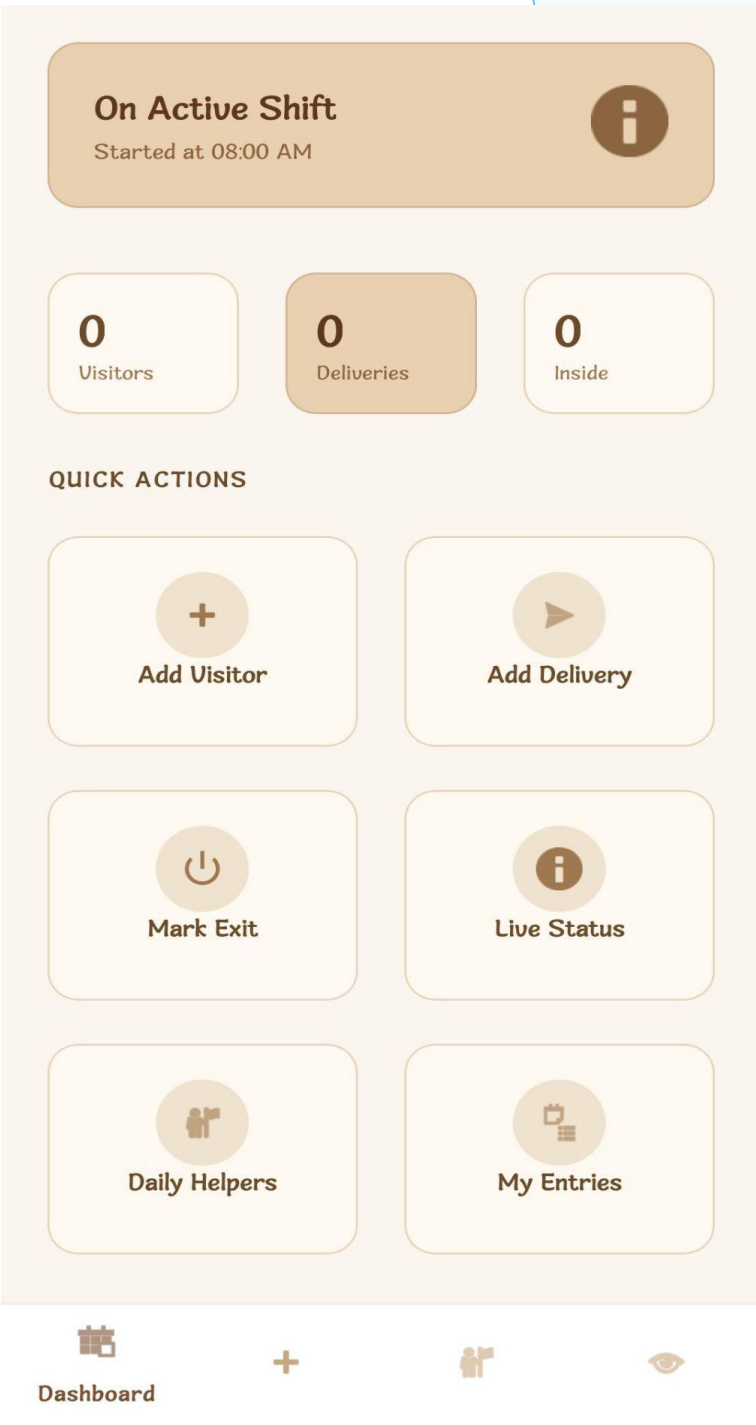
Pending
Management
Manage Users
Visitor Logs
Delivery Logs
Approvals
Reports
Complaints
Dashboard


Good Morning!
Flat A-102 • Owner
0
Visitors
0 pending approvals
0
Deliveries
Expected today
1 active issues
Complaints
Active issues
1
Helpers
Manage daily
Unpaid
Maintenance
View & Pay Bills
Quick Actions
Contact Guard
Notice Board
Complaint
Home
Requests
History
Helpers
Profile


On Active Shift
Started at 08:00 AM
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Visitors
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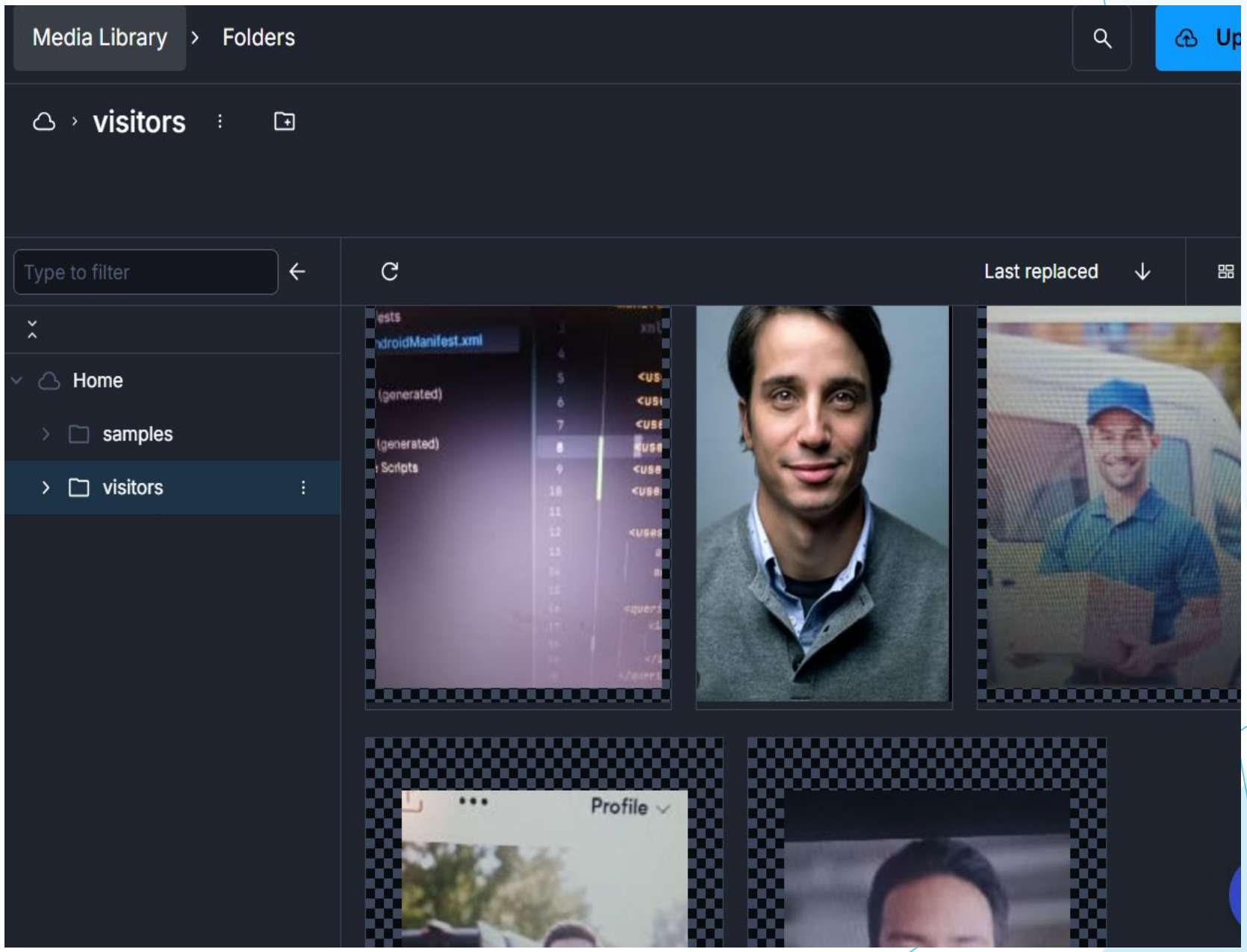
Deliveries
0
Inside
QUICK ACTIONS
Add Visitor
Add Delivery
Mark Exit
Live Status
Daily Helpers
My Entries
Dashboard


Media Library
Folders
visitors
Type to filter
Last replaced
Home
samples
visitors
Profile

securiodb ▾ Cloud Firestore › Database (default)

Panel view Query builder

› dailyHelpers › lYWhXhbKZASz...

More in Google Cloud

(default)

+ Start collection

- bills
- complaints
- dailyHelpers
- notifications
- users
- visitors

dailyHelpers

+ Add document

- AkiVvGkpEKSkktHY4m6I
- fyaBnTITze6G33rRqRGX
- lYWhXhbKZASz2RtqBJa1

lYWhXhbKZASz2RtqBJa1

+ Start collection

+ Add field

```
addedOn: 1774953602145
entryTime: "8"
flatNumber: "A-101"
helperName: "Neha Kakkar"
helperPhone: "9863512475"
helperType: "Gardener"
isActive: true
ownerUid: "u8ufMgVZ6Vh8PLqS7Lr4BbkEBO23"
schedule
  0 "Monday"
  1 "Wednesday"
  2 "Friday"
```

securiodb

Authentication

Users | Sign-in method | Templates | Usage | Settings | Extensions

The following Authentication features will stop working when Firebase Dynamic Links shuts down soon: email link authentication for mobile apps, as well as Cordova OAuth support for web apps.

Search by email address, phone number, or user UID

Add user

Identifier	Providers	Created	Signed In	User UID
neha11@gmail.com		Mar 31, 2026	Apr 1, 2026	5LbFXhTDPYMlKAcCWZPJXIh...
shravanibhilare.genai@...		Mar 31, 2026	Mar 31, 2026	k48jJZdRybM8t3e7Bu0MwBQ...
shravu@gmail.com		Mar 31, 2026	Mar 31, 2026	Tm96SlOKDvRPvW1UauQnpo...
rony@gmail.com		Mar 30, 2026	Apr 1, 2026	ry5vL5Qy15hmchk4ALVWccR...
abcda@gmail.com		Mar 30, 2026	Mar 30, 2026	zHo2R1gUPldkjQ5wzUPT9lm1...
shravanibhilare25@gm...		Mar 30, 2026	Mar 31, 2026	YT8yjZRTaAWoBa7mai83QDE...
g@gmail.com		Mar 30, 2026	Apr 1, 2026	du9hkI13jJNpQ1kwQe8Ogvmk...

Thank You...!!